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Numerical investigation of hydrogen absorption in a stackable metal hydride reactor utilizing compartmentalization

Purushothama Chippar [a,*], *Swaraj D. Lewis* [a], *Sudhir Rai* [b], *Amit Sircar* [b]

[a] *St. Joseph Engineering College, Vamanjoor, Mangaluru 575028, Karnataka, India*

[b] *Institute for Plasma Research, Gandhinagar 382428, Gujarat, India*

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ABSTRACT

In this paper, a three-dimensional model for hydrogen absorption in a metal alloy has been developed, validated against the experimental data in the literature, and then applied to a novel design for a hydrogen storage unit. The proposed design is similar to the fuel cell stack, but here the Membrane Electrode Assembly (MEA) has been replaced by a metal hydride (MH) reactor placed between the flow-field plates. These are stacked together to achieve the required amount of hydrogen storage. The flow-field plates have channels engraved on one side for hydrogen supply and on the other, for coolant/heating medium. It is known that the effectiveness of a hydrogen storage unit is directly related to its heat transfer area, and therefore, the choice of its geometry is very important. The larger the size, the more the resistance to heat transfer. Although, the internal tubular heat exchangers have proven to be effective in heat transfer, they pose severe challenges such as cooling/heating medium leakage due to tube erosion, stresses generated, etc. and they displace the active metal hydride from the tank. The present stacked MH reactor configuration helps to overcome these challenges by stacking small MH reactors together and there is no chance of the cooling/heating medium leaking into the metal hydride. Numerical simulations were performed to investigate the effect of coolant flow rate and percentage of flow-field plate rib area exposed to the MH reactor on temperature evolution and the amount of hydrogen stored. Further, a detailed study was carried out to understand the effect of compartmentalization of the MH reactor on temperature distribution. The results revealed that compartmentalization substantially helps to uniformly distribute the temperature in the metal bed, which is very important to maintain uniform utilization of the metal powder. Consequently, the uniform metal powder density for repeated absorption-desorption cycles without significant loss of its hydrogen storage capabilities.

* *Corresponding author.*
E-mail address: purushothama@sjec.ac.in (P. Chippar).

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Nomenclature

C_a	absorption rate constant, s^{-1}
C_P	specific heat, kJ.(kg K)$^{-1}$
E_a	activation Energy, J.mol^{-1}
$\frac{H}{M}$	hydrogen to metal atomic ratio
ΔH	reaction heat of formation, J.kg^{-1}
h	heat transfer coefficient, W.(m^{2}K)$^{-1}$
K	permeability, m^2
M	molecular weight, kg.mol^{-1}
P	permeability, m^2
R	universal gas constant, 8.314 J.(mol K)$^{-1}$
T	temperature, K
$\vec{u}$	velocity vector. m.s^{-1}
Greeks	
ε	porosity
ρ	density, kg.m^{-3}
k	thermal conductivity, W.(mK)$^{-1}$
μ	dynamic viscosity, kg.(m s)$^{-1}$
Superscripts	
g	gas phase
s	solid phase
sat	saturation value
Subscripts	
eq	equilibrium
H_2	hydrogen
M	metal
ref	reference value
0	initial value

Introduction

Hydrogen is considered as a promising energy carrier in the foreseeable future due to its high energy density on mass basis and clean combustion without producing harmful pollutants. However, its low energy density per unit volume under normal pressure and temperature has made its storage and transportation a major bottleneck towards the hydrogen economy. There have been numerous research and developments with regard to the conventional methods to store hydrogen such as compression and liquefaction, however, the critical technically challenging barriers still remain. Solid state storage of hydrogen in metal, alloy, and intermetallic hydrides has gained much attention recently and significant research is ongoing to improve not only gravimetric and volumetric capacities also the hydrogen absorption and desorption reaction thermodynamics and kinetics [1]. Two key processes, which determine the absorption and desorption performance of solid state storage of hydrogen are mass and heat transport. The absorption and desorption of hydrogen are associated with considerable amount of heat release and uptake, respectively. The poor thermal conductivity of raw metal powder requires an effective heat management system. Therefore, the design of a metal hydride (MH) reactor with an effective heat exchanger to achieve rapid hydrogen charging and discharging rates is very important, and this has generated a lot of interest among researchers.

Mellouli et al. [2] and Souahila et al. [3] experimentally studied the heat transfer characteristics in an MH reactor equipped with a spiral heat exchanger. Their results emphasized the importance of successful heat removal from the metal bed and indicated the significance of the selection of operational parameters to improve the charging time of the reactor. They also tested this spiral heat exchanger with copper fin coils inserted between the turns [4]. The results showed considerable improvement in the charging and discharging times of the MH reactor. A reactor [5] similar to the above references was tested with a concentric heat exchanger equipped with stainless steel fins, and it was noticed that the design of the heat exchanger, filters, and geometrical distribution of the alloy powders are important issues related to the improved performance of hydrogen storing devices. Linder et al. [6] used a capillary tube bundle reaction bed proposed by Willers and Groll [7] and experimentally investigated the main influencing parameters on the dynamics of an MH bed containing $LnNi_{4.91}Sn_{0.15}$ as a hydride material. They compared the dynamics of the capillary tube bundle reaction bed with the intrinsic desorption kinetics of $LnNi_{4.91}Sn_{0.15}$ and concluded that the intrinsic reaction kinetics limited the achievable desorption rate. Visaria et al. [8] conducted experiments on a coiled tube heat exchanger utilizing $Ti_{1.1}CrMn$ up to 280 bar and found that the distance of the MH particles from the coolant tube played a key role in determining the sorption rate.

In addition, several attempts were made to use a heat pipe to enhance the sorption rate in MH tanks [9–11]. Many research works have been devoted to improving the thermal conductivity in an MH reactor by insertion of metal foams [12], integration of copper wire net structure [13], and micro encapsulated metal hydride compact [14].

Simultaneously, a number of numerical analyses were carried out to further delineate mass transport, sorption reaction, and heat transfer mechanisms in MH reactors [15–24]. Timothée et al. [15] conducted experimental and numerical analysis of hydriding and dehydriding with 100 g $Ti_{1.1}MnCr$ in a reactor fitted with a finned tube-in-tube heat exchanger under different operating pressures and coolant flow rates. The lack of agreement in the temperature profiles between the experimental and numerical results indicated the importance of considering proper metal properties, particularly thermal conductivity as a function of the concentration of hydrogen in the numerical models. Shahrzad et al. [16] developed an MH reactor model with finned coolant tube heat exchanger. Special emphasis was placed on using effective thermal conductivity and equilibrium pressure as a function of temperature and concentration of hydrogen. Mellouli et al. [17] conducted experimental and numerical analyses to investigate the change in specific volume of a $LaNi_5$ reactor during absorption and desorption, which could lead to variation in the thermo-physical properties of the metal. Significant change in the specific volume was observed with approximately 18 vol% and 8 vol% during charging and discharging, respectively. However, the effect

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of hydrogen concentration on thermal conductivity is not yet clear. The studies [25–28] shown increase in the effective thermal conductivity, whereas an experimental study by Jörg and Jobst [29] shown decrease in the effective thermal conductivity with increasing reacted fraction. A study by Kempf and Martin [30] found that hydriding of the powders does not significantly change the effective thermal conductivity.

Muthukumar et al. [31] performed thermal and performance analysis of an industrial-scale MH container with various coolant tube configurations and with/without outer cooling jacket. They inferred that in order to achieve the minimum absorption time and manufacturing complexity, it is very important to optimize the geometry and operating parameters of the MH container. Haneul et al. [32] numerically accessed the hydrogen absorption phenomena in a thin double-layered annulus MH bed. Their results showed uniform temperature and hydrogen-to-metal (H/M) ratio, indicating the effective heat transfer from the bed to the ambient coolant. Chung et al. [33] developed a numerical model for simulating the thermo-fluid behaviors of an MH tank embedding heat pipes, which clearly showed the effects of using heat pipes on the absorption and desorption rates. Haidi et al. [34] proposed a spiral mini channel reactor and numerically investigated the effects of structural parameters such as minor radius, major radius, spiral tube amount, axial pitch, and operating conditions on the absorption rate. Few attempts were made to use the phase change materials to store the latent heat released during the absorption and to release the same during the desorption process [35–37]. Although, the internal tubular heat exchangers improved the hydrogen sorption capability significantly, they also posed serious technical concerns. One of them is the leakage of the coolant/heating medium from the tube due to its erosion or stresses generated by the metal powder, which can contaminate the metal powder [9,38].

In this paper, we present a novel MH reactor stack configuration, where small sized MH reactors stacked together enable quick heat transfer between the cooling/heating medium and metal bed. The present design of the MH reactor completely avoids any leakage of the cooling/heating medium into the metal alloys.

Mathematical model

The present three-dimensional hydrogen absorption model for $LaNi_5$-H_2 system is governed by mass, momentum, and energy conservation equations along with various constitutive relations. The specific assumptions made in this model include the following: (1) the gas phase obeys the ideal gas law because the pressure within the MH bed is moderate and the temperature is much above the critical temperature; (2) the metal bed is considered as an isotropic and homogeneous porous media and characterized by effective porosity, permeability, and thermal conductivity; (3) local thermal equilibrium between the solid and gas phases, and (4) the thermo-physical properties in the bed are independent of temperature, pressure, and concentration.

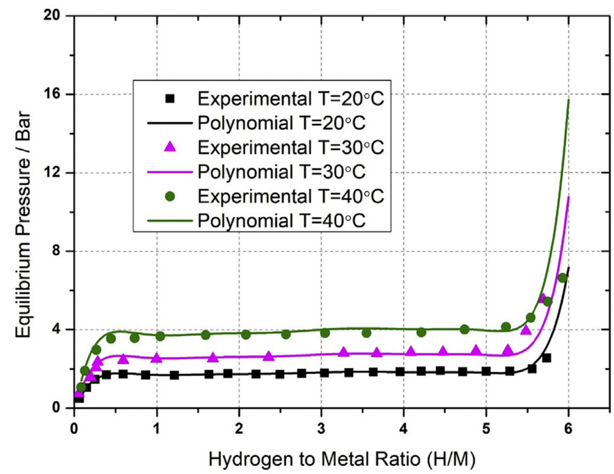

Fig. 1 – Polynomial fit of equilibrium pressure as a function of hydrogen-to-metal (H/M) ratio and temperature calculated by Eq. (5).

Mass balance

The continuity equation solved for hydrogen balance in the MH bed is expressed as follows:

For hydrogen:

$$\frac{\partial \varepsilon \rho^g}{\partial t} + \nabla \cdot (\rho^g \vec{u}) = -\dot{m} \tag{1}$$

where, ρ^g is the gas density described by the ideal gas law as follows:

$$\rho^g = \frac{P^g M^g}{RT} \tag{2}$$

For metal:

$$\frac{(1-\varepsilon)\partial \rho^s}{\partial t} = \dot{m} \tag{3}$$

In Eqs. (1) and (3), $\dot{m}$ represents the hydrogen absorption rate per unit volume and is determined by the reaction kinetics of $LaNi_5$ as follows [39]:

$$\dot{m} = C_a \exp\left(-\frac{E_a}{RT}\right) \cdot \ln\left(\frac{P^g}{P_{eq}}\right)\left(\rho^s_{sat} - \rho^s\right) \tag{4}$$

where, P_{eq} denotes the equilibrium pressure in the MH bed during the absorption process and is determined as a function of temperature and H/M as follows [40]:

$$P_{eq}(bar) = f\left(\frac{H}{M}\right) \cdot \exp\left(\frac{\Delta H}{R_g}\left(\frac{1}{T} - \frac{1}{T_{ref}}\right)\right) \tag{5}$$

Here, $f\left(\frac{H}{M}\right)$ is the polynomial function of the order 9 and represents the equilibrium pressure at reference temperature T_{ref}. The polynomial function coefficients listed below are derived from the experimental P-C-T curves for three reference temperatures from Jemni et al. [41]. The experimental and polynomial fitted equilibrium pressure evolution for $LaNi_5$-H_2 system as a function of temperature and H/M ratio are shown in Fig. 1.

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$$f\left(\frac{H}{M}\right) = a_0 + \sum_{n=1}^{9} a_n \left(\frac{H}{M}\right)^n$$
$$a_0 = -0.23691; a_1 = 16.9663; a_2 = -38.94368; a_3 = 46.25223; a_4 = -32.09352;$$
$$a_5 = 13.7198; a_6 = -3.65573; a_7 = 0.59087; a_8 = -0.05297; a_9 = 0.0020197 \tag{6}$$

Momentum balance

$$\frac{1}{\varepsilon}\frac{\partial \rho^g \vec{u}}{\partial t} + \frac{1}{\varepsilon^2}\nabla\cdot(\rho^g \vec{u}\vec{u}) = -\nabla p + \nabla\cdot\tau + S_u \tag{7}$$

The momentum source term in the metal bed is calculated using Darcy's law - $S_u = -\frac{\mu}{K}\vec{u}$, while in the other regions - $S_u = 0$.

Energy balance

Assuming local thermal equilibrium between the hydride and hydrogen, the standard energy equations available in ANSYS-FLUENT are used to solve the temperature field in non-porous and porous mediums.

$$\text{Non}-\text{porous medium}$$
$$\frac{\partial(\rho^g E^g)}{\partial t} + \nabla\cdot(\vec{u}(\rho^g E^g + P)) = -\nabla(k^g \nabla T + \tau.\vec{u}) \tag{8}$$

$$\text{Porous medium}$$
$$\frac{\partial(\varepsilon\rho^g E^g + (1-\varepsilon)\rho^s E^s)}{\partial t} + \nabla\cdot(\vec{u}(\rho^g E^g + P)) = -\nabla\left[k_{eff}\nabla T + \tau.\vec{u}\right] - \dot{m}\left[\Delta H + T\left(C_P^g - C_P^s\right)\right] \tag{9}$$

where, E^g,E^s, and k_{eff} are total fluid energy, total solid medium energy, and effective thermal conductivity of the medium, respectively. The second term at the right hand side of Eq. (9) represents the energy source term during exothermic hydrogen absorption. Effective thermal conductivity can be expressed as a function of porosity-weighted function of hydrogen gas and solid medium thermal conductivities as below:

$$k_{eff} = \varepsilon k_{H_2} + (1-\varepsilon)k_M \tag{10}$$

Initial/boundary conditions and numerical implementation

Initially, the metal hydride density, temperature, and pressure in the MH bed are assumed to be uniform. A constant hydrogen supply pressure and temperature is maintained at the inlet port. Therefore,

$$\rho^s = \rho_0^s; T = T_0; P^g = P_0; P_{in} = 10\ atm; T_{in} = 293.15K \tag{11}$$

where, ρ_0^s represents the density of the initial hydrogen-empty metal bed. No-slip velocity and no-flux conditions are applied as boundary conditions to the side walls of the tank. The model is numerically implemented into a commercial Computational Fluid Dynamics (CFD) package, ANSYS-FLUENT 17.2 using its user-subroutine capability. A grid independence test is performed by analyzing the effect of grid size on the evolution of average bed temperature, and the number of meshes is optimized. For this purpose, a cylindrical MH reactor (used in validation of the numerical model) and stackable MH reactor (flow-field plate 1, see Fig. 7) are considered. The grid sizes are varied from coarse, medium, fine to ultrafine and the corresponding number of elements are provided in Table 1. As illustrated in Fig. 2a and b, the results of average bed temperature for cylindrical and stackable MH reactor obtained from fine and ultrafine grid match closely. Therefore, fine grid is considered for further analyses. The qualitative convergence of mass, momentum, and energy equations are checked using the residual history set for 10^{-6}. In addition, the quantitative convergences of mass and energy equations are checked via monitoring their imbalance.

Results and discussion

Model validation

To validate the present model, we used the experimental results from Jemni et al. [41]. They used a cylindrical tank of 50 mm diameter and 80 mm height filled with 422 g of $LaNi_5$ powder of 63 μm size. The tank was placed in a water bath and the temperature was set to 20 °C. Our model was of similar dimension and was filled with the same amount of metal powder to a tank depth of 60 mm. Again, based on their work, the convective heat transfer coefficient and temperature external to the tank were specified as 1652 W/mK and 293.15 K, respectively. The heat flux from the external wall of the tank via convection was calculated as follows:

$$\kappa_{wall}\nabla T|_{wall} = h(T_{wall} - T_\infty) \tag{12}$$

Since we did not consider the thickness of the tank in our simulation, κ_{wall} corresponds to the thermal conductivity of the cells near to the tank walls. The operating conditions and thermo-physical properties of the metal powder used are listed in Table 2. The temperature in the MH bed was monitored at three locations, A, B, and C at a height of 35 mm, 25 mm, and 15 mm, respectively, and at a radius of 15 mm similar to the experimental work. Fig. 3 shows the comparison of the simulation and experimental results for the temperature at the three locations in the MH bed. The predicted temperature agreed closely with the experimental data. Both the experimental and simulation results showed that the temperature in the MH bed increased sharply during the initial stages due to the fast rate of hydrogen absorption driven by low equilibrium pressure, which is a function of both temperature and H/M ratio (Eq. (5)), as well as the availability of the free metallic

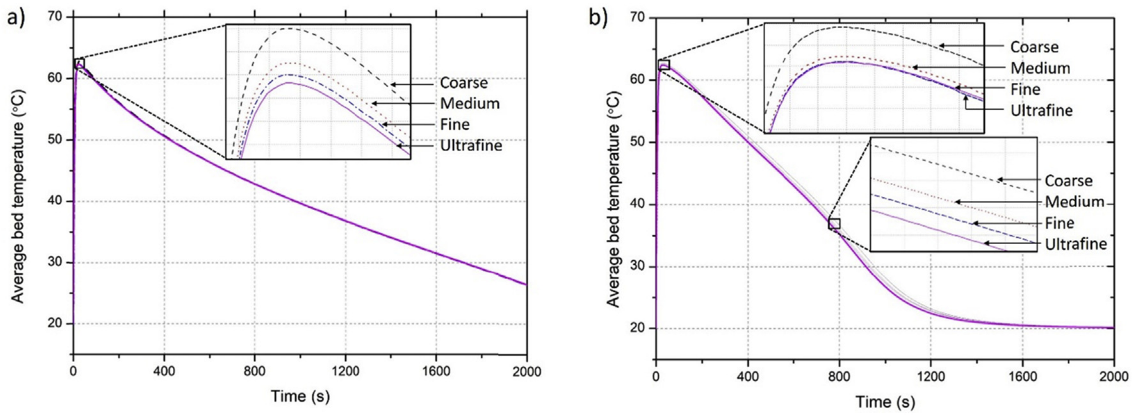

Fig. 2 – Grid independency analyses: a) for cylindrical MH reactor and b) stackable MH reactor.

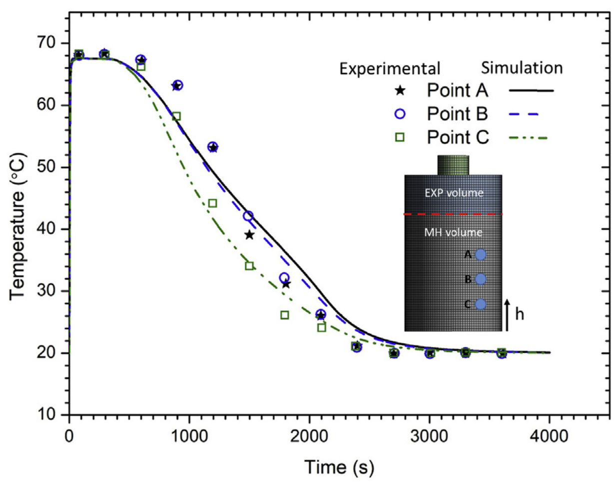

Fig. 3 – Simulation and experimental temperature evolution at three monitoring points (A, B and C) in the MH bed. (Point A: r = 15 mm, h = 35 mm, B: r = 15 mm, h = 25 mm, C: r = 15 mm, h = 15 mm).

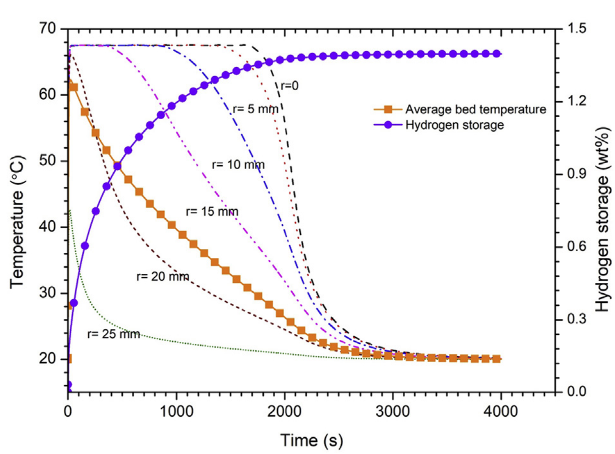

Fig. 4 – Temperature at different radial locations, average temperature, and hydrogen storage evolution in the MH bed.

surfaces. The temperature at the three locations in the MH bed reached a maximum value of 68 °C in 18 s, where the equilibrium pressure was observed to be very close to the supply pressure of 10 atm. From 18 s to 400 s, the temperature at these three locations was observed to be around 68 °C, which is an indication of the balance effect between the high equilibrium pressure trying to retard the absorption process and the available free metal surface, which promotes the absorption process. As the time proceeds, the temperature in the MH bed gradually declines with the decrease in the available free metallic surfaces, and finally, approaches the temperature of the water bath at 3000 s. It should be noted that at points A, B, and C, whose radius is 15 mm, the high equilibrium pressure counters the effect of the availability of the free metallic surfaces on the absorption process. Therefore, a pause in temperature is observed from 18 s to 400 s. Due to low thermal conductivity of the metal bed, the equilibrium pressure plays a significant role in retarding the hydrogen absorption at the core of the tank. However, near the tank wall region, due to effective convective cooling, the sharp rise in temperature will have a momentarily pause, followed by gradual decline. This is evident from Fig. 4, which shows the temperature evolution at different radii and height 30 mm, average temperature, and overall hydrogen content in the MH bed. The simulation shows that the hydrogen absorption approaches 90% of saturation in 1200 s and the metal bed fully saturates in 3000 s.

Model implementation to a stackable MH reactor

The MH absorption model was further used to study the feasibility of the stack reactor configuration, similar to a fuel cell stack, for hydrogen storage. In the present study, the Membrane Electrode Assembly (MEA) of the fuel cell stack was replaced by a hydrogen reactor. The main objective of stacking the hydrogen reactors was to provide multiple smaller hydrogen storage volumes for effective heat transfer into or out of the bed. This will also ease customizing the size of the reactor based on the amount of hydrogen to be stored. There have been several research works on the compartmentalization of the reactor to improve heat transfer; however, the authors believed that the stacking of hydrogen reactors used in this study, similar to the fuel cell stack, is a novel concept.

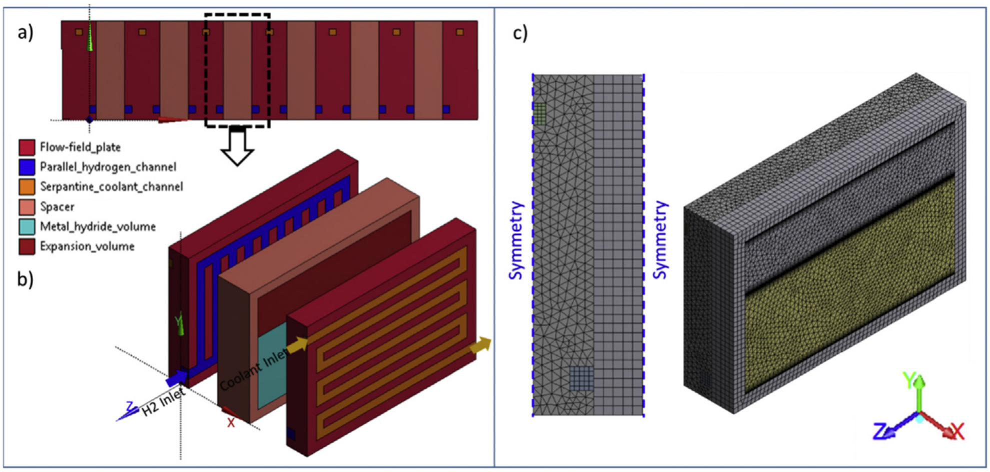

Fig. 5 – **a) Stack reactor configuration for hydrogen storage, b) unit reactor, and c) computational domain.**

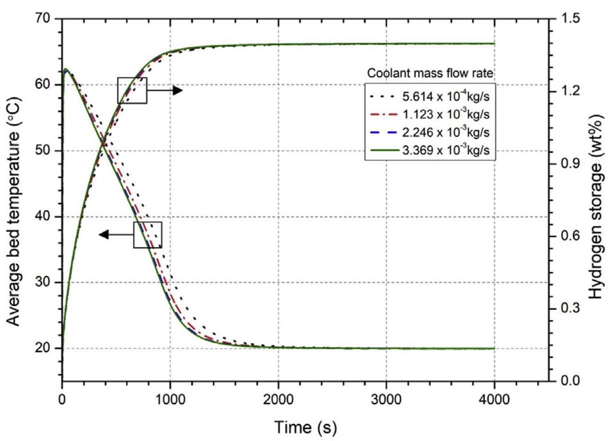

Fig. 6 – **Average bed temperature and hydrogen storage evolution curves under different coolant flow rates (Coolant inlet temperature is 20 °C).**

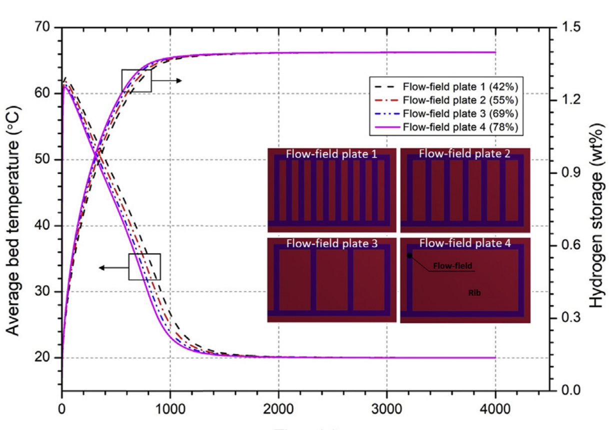

Fig. 7 – **Average bed temperature and hydrogen storage evolution curves for four designs of flow-field plates having different rib area exposed to the metal bed.**

Table 1 – **Grid sizes based on the element size.**

Grid Name	Minimum element size (mm)	Maximum element size (mm)	No. of elements
Cylindrical MH reactor			
Coarse	0.0197	3.93	13,510
Medium	0.00155	3.09	26,578
Fine	0.00139	2.77	35,588
Ultrafine	0.00126	2.51	48,872
Stackable MH reactor			
Coarse	0.0238	4.75	111,446
Medium	0.0205	4.09	158,125
Fine	0.0187	3.74	198,285
Ultrafine	0.0168	3.36	248,003

Table 2 – **Operating conditions and thermo-physical properties.**

Description	Value	Ref.
Initial & inlet temperature, T_0&T_{in}	293.15 K	[41]
Inlet pressure, P_{in}	10 atm	[41]
Reference temperature, T_{ref}	293.15 K	[41]
Activation energy, E_a	21,179.6 J mol^{-1} K^{-1}	[42]
Heat of formation, ΔH	-1.54×10^7 J kg^{-1}	[42]
Absorption rate coefficient, C_a	59.187 s^{-1}	[42]
Specific heat of hydrogen, C_{P,H_2}	14,890 J mol^{-1} K^{-1}	[42]
Specific heat of metal, $C_{P,M}$	419 J mol^{-1} K^{-1}	[42]
Thermal conductivity of hydrogen, κ_{H_2}	0.167 W m^{-1} K^{-1}	[17]
Thermal conductivity of metal, κ_M	1.5 W m^{-1} K^{-1}	[41]
Porosity of metal bed, ε	0.63	[41]
Hydrogen-free metal density, ρ_0^s	8175 kg m^{-3}	[33]
Permeability of the metal bed, K	10^{-8} m^2	[33]

Fig. 5 shows the schematic diagrams of the stack reactor configuration (Fig. 5a), unit reactor (Fig. 5b), and the computational domain for the model implementation (Fig. 5c). Two flow-field plates with parallel channel configuration deliver hydrogen to the reactor from the manifold (manifold is not shown). The flow-field plates also have single-pass serpentine channel on the other side for passing the coolant. A metallic spacer with leakage-proof gaskets (not shown) is provided between the flow-field plates. It is also possible to weld the

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spacer to the flow-field plates. The total volume of the unit reactor is 330.75 cm^3 (10.5 cm (l) × 4.5 cm (w) 7.0 cm (h)). The metal bed and expansion volumes are 74.1 cm^3 (9.5 cm (l) × 2.0 cm (w) 3.9 cm (h)) and 39.9 cm^3 (9.5 cm (l) × 2.0 cm (w) 2.1 cm (h), respectively. The hydrogen and coolant supply channels had width of 5.0 mm and 4.5 mm, respectively; all the channels had a depth of 5.0 mm. The size of the computational domain is reduced by taking the geometrical symmetry of the unit reactor. The heat generated during absorption in the reactor is solely removed by the coolant in the coolant channel. Adiabatic condition is applied to the other external surfaces of the computational cells for the temperature calculations, except for the hydrogen inlet and the coolant channel inlet and outlet. The flow-field and cover plates are impermeable for hydrogen. The following sections discuss in detail the effect of the coolant flow rate, rib exposure area to the metal bed, and compartmentalization of the unit reactor on the bed temperature and hydrogen storage.

Effect of coolant flow rate

The influence of the coolant flow rate on bed temperature and hydrogen storage was investigated by varying the coolant flow rates. Fig. 6 shows the average bed temperature and hydrogen storage (wt%) for four different coolant flow rates. Higher coolant flow rate decreases the bed temperature, improves the absorption rate, and therefore, higher the hydrogen storage. However, the effect of coolant flow rate on bed temperature and hydrogen storage diminishes with increasing flow rate, indicating that heat transfer is strongly controlled by the thermal conductivity of the medium in the bed. During the initial stages of absorption, the exothermic heat is very high and the increase in arbitrarily chosen coolant flow rates in this simulation does not improve the heat transfer significantly. As the absorption proceeds, the effect of coolant flow rate on bed temperature and hydrogen storage is clearly visible in the figure.

Effect of rib exposure area to the metal bed

To examine and discuss the effects of percentage of rib area exposed to the metal bed on temperature and hydrogen storage, four designs of the flow-field plates were considered with fixed coolant flow rate of 3.369×10^{-3} kg/s. Fig. 7 shows the average temperature and hydrogen storage (wt%) in the bed with the four flow-field plates. The bed temperature reduces, and consequently, the hydrogen storage improves with the increase in rib area exposed to the metal bed due to the enhanced heat transfer. Similar to the coolant flow effect discussed in the previous section, the improvement in heat transfer declines with the increase in rib area exposed to the metal bed. Again, it shows that the heat transfer limiting step is the thermal conductivity of the medium in the bed. Similar to the previous simulation results, the rib area exposed to the metal bed becomes significant as the hydrogen absorption proceeds until the bed temperature reaches the coolant temperature.

Compartmentalization of the reactor

Two generally recognized difficulties with metal hydrides are low thermal conductivity and non-uniform metal powder density in the bed after repeated cycles of absorption and desorption. Since the virgin metal alloy has low thermal conductivity, and in its hydride state is somewhat analogous to metal oxides, it behaves like an insulating material. As metal alloys expand and contract during the absorption and desorption of hydrogen, respectively, the several repeated cycles of these processes result in comminution of the alloys into successively finer grains. These fine grains settle in the lower regions of the tank, leading to highly packed powder density, and finally, result in localized excessive heating during absorption. This produces a great amount of local strains and can deform/rupture the tank. Therefore, an effective heat transfer within the tank is of essence to improve hydrogen storage, and also to maintain uniform powder density in the metal bed.

The compartmentalization of the reactor will help to transfer the heat into and out of the bed effectively, thereby leading to uniform temperature distribution, and eventually to uniform utilization of the metal powder and uniform powder density in the bed. To study the effect of compartmentalization on the present reactor concept, we defined 5 simulation cases. The flow-field plate design 3 with coolant flow rate 3.369×10^{-3} kg/s was used in all the case studies. The 5 simulation cases are: Case 1-without reactor compartmentalization, Case 2-with three vertical compartments, Case 3-with 3 vertical and 2 horizontal compartments, Cases 4 and 5 are modified geometries of Cases 2 and 3, respectively, in which the compensative metal bed volume is considered from the expansion volume for the volume occupied by the compartment structures. The vertical and horizontal compartments had a thickness of 2.5 mm. Fig. 8 shows the average temperature and hydrogen storage in the metal bed for the above 5 simulation cases. Comparing Cases 1 and 2, significant improvement in heat transfer out of the bed was observed with vertical compartmentalization and the 90% saturation time decreased from 618 s to 492 s, thus reducing the charging time by 20%. In Case 3, which has both vertical and horizontal compartments, further improvement in heat transfer and hydrogen storage in the bed was noticed. The time taken for 90% of saturation was 415 s, reducing the charging time by 33% from Case 1 and 16% from Case 2. It should be noted that the compartment structures reduce the volume available for the metal hydride, which can

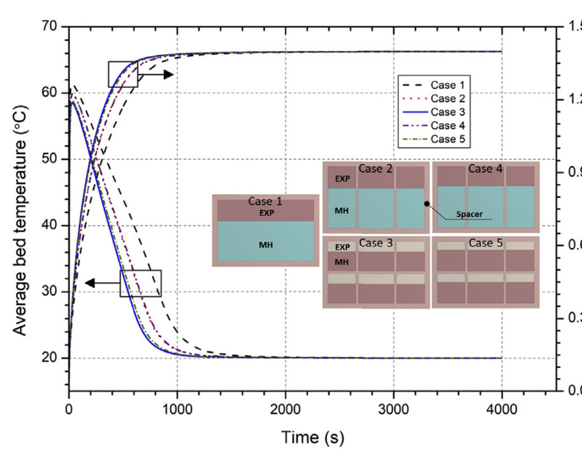

Fig. 8 – Average bed temperature and hydrogen storage evolution in the metal bed for Cases 1-5.

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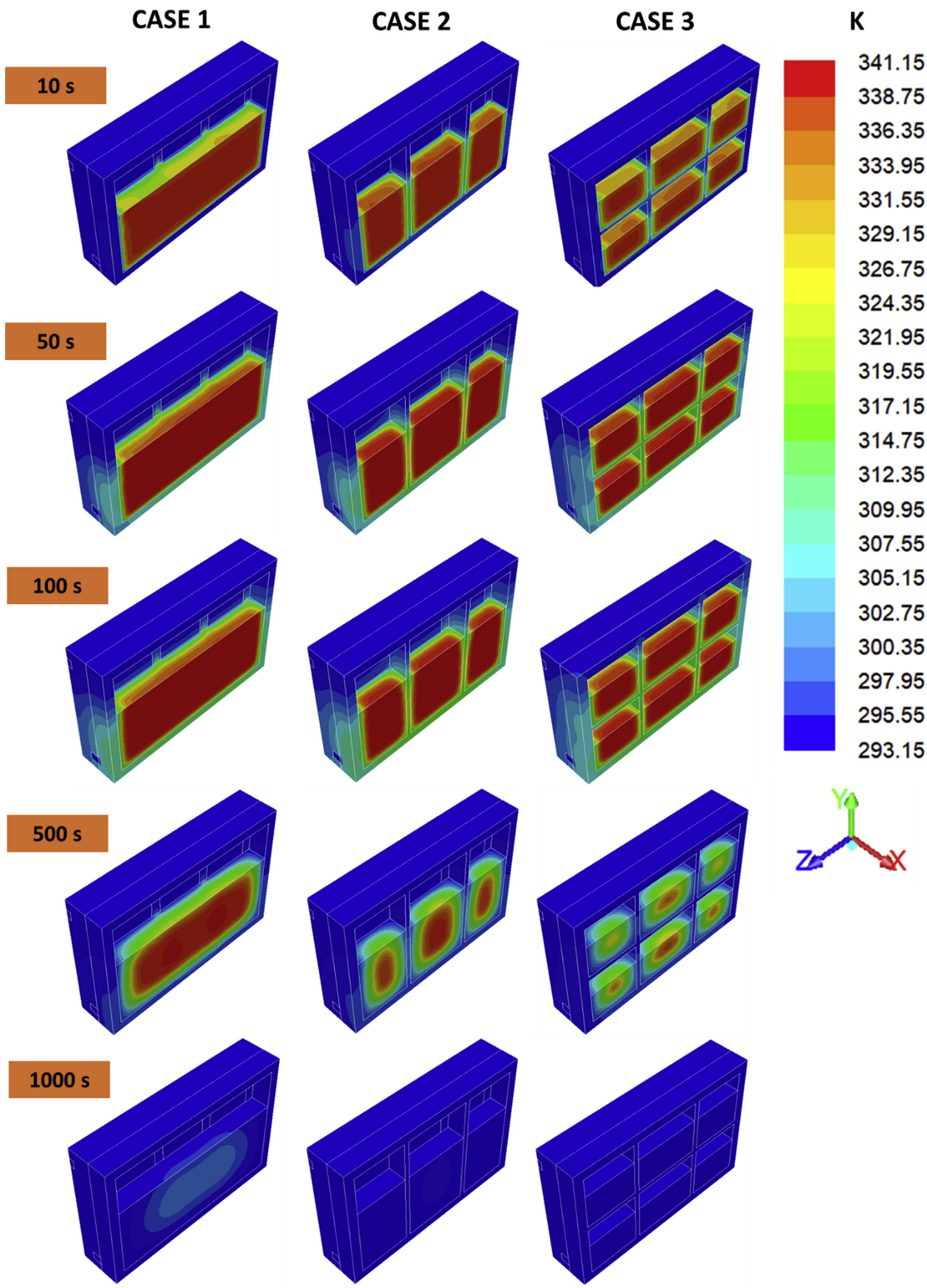

Fig. 9 – Contours of local temperature for Cases 1, 2, and 3 at different intervals of time.

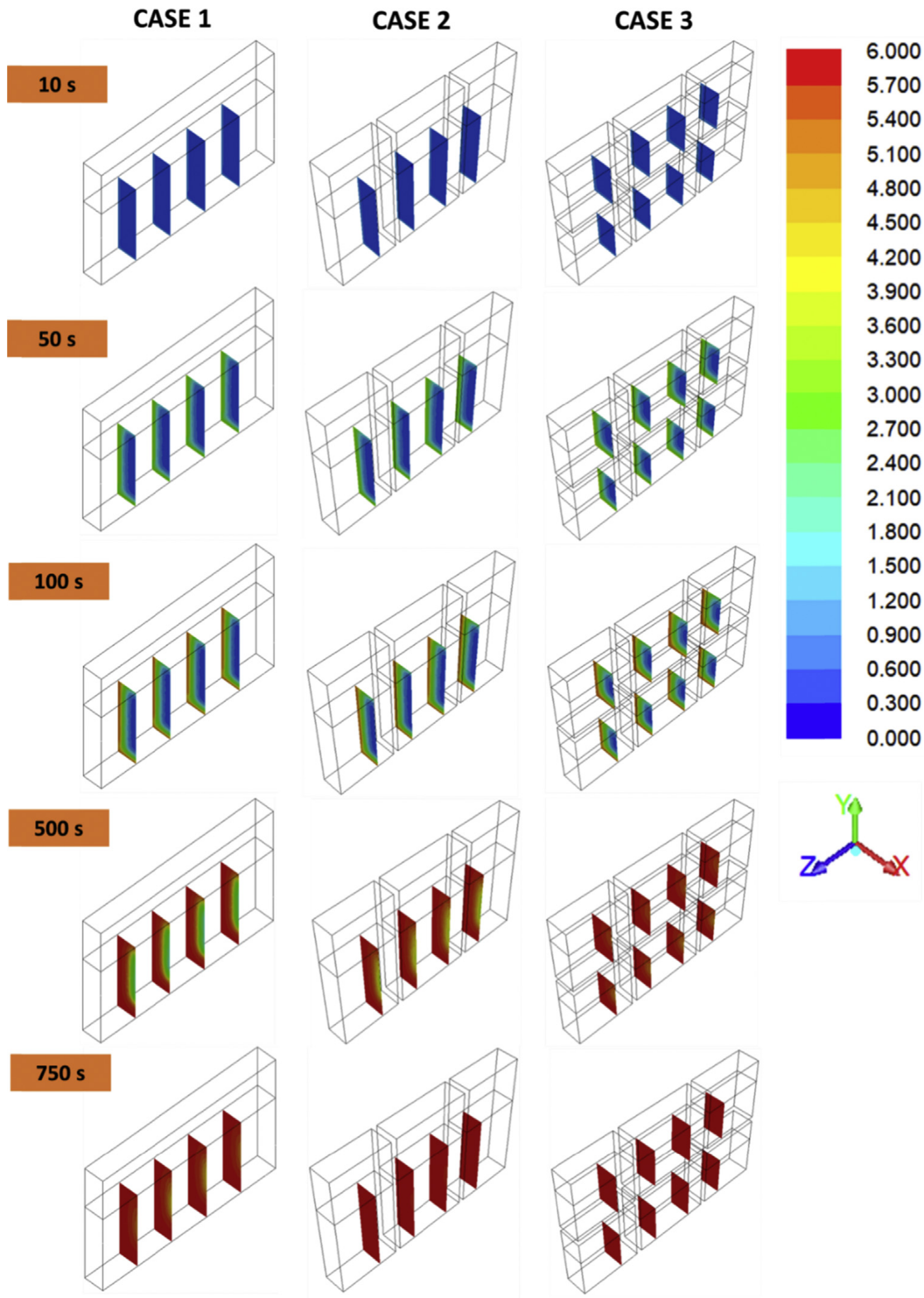

Fig. 10 – Contours of local H/M ratio for Cases 1, 2, and 3 at different intervals of time.

compromise the hydrogen storage. In our simulations, the volume occupied by the compartmentalization structures in the metal bed was 5.2% and 9.2% of the total metal bed volume for Case 2 and Case 3, respectively. In order to understand this effect on the temperature and hydrogen storage in the bed, two additional simulations were carried out with Cases 4 and 5, where the compensation metal bed volumes were considered from the expansion volume. As shown in Fig. 8, comparing Case 2 with Case 4 and Case 3 with Case 5, it was clear that the effect of volume occupied by the compartmentalization structure is negligible on both average temperature and hydrogen storage in the metal bed for the reactor geometry considered in the present work.

To analyze the effects of compartmentalization more precisely, detailed multidimensional distributions of temperature and H/M ratio are provided in Figs. 9 and 10, respectively. Fig. 9 shows the temperature distribution for Cases 2 and 3 in comparison with Case 1 in all the components of the reactor, except expansion volume, at various time intervals. The temperature in the metal bed monotonically increases during the initial stages for all the three cases almost at the same rate due to fast hydrogen absorption. As the absorption proceeds, the compartmentalization plays a significant role on temperature distribution due to improved heat transfer rate. At 500 s, Case 1 had a large area with temperature above 338 K, whereas in Cases 2 and 3, only a small region in the centre of the metal bed showed a high temperature. Comparing Cases 2 and 3, it is clear that the presence of both, horizontal and vertical compartments, enhance the heat transfer out of the metal bed. The entire metal bed in Cases 2 and 3 reached coolant temperature at 1000 s, whereas the central region in Case 1 was still around 302 K. The H/M ratio distribution in the metal bed shown in Fig. 10 exhibited the opposite of that temperature. The hydrogen was absorbed at a faster rate near the spacer and compartment structures due to greater heat transfer efficacy compared with the other regions. At 500 s, the centre of the metal bed in Case 1 was less than 50% saturated and in Cases 2 and 3, nearly and almost saturated, respectively, whereas at 1000 s, a small region was still unsaturated.

Summary and conclusions

A three-dimensional numerical model was developed to study the heat and mass transfer during hydrogen absorption in an MH tank and validated against the experimental data of Jemni et al. [41]. The validated numerical model was used to investigate the feasibility of stack reactor configuration for hydrogen storage and the enhancing of heat transfer by compartmentalization of the reactor. To simplify the simulation, a unit reactor was considered with a symmetrical boundary. The heat generated in the metal bed during the exothermic absorption of hydrogen was carried out solely by the coolant running in the flow-field plate. A parametric study was performed to investigate the role of various factors such as coolant flow rate, flow-field design, and compartmentalization of the reactor that can influence the absorption process. The numerical results showed that increasing the coolant flow rate had a negligible influence on hydrogen absorption during the initial stage; however, it played an important role as the absorption proceeded. The simulations on the effect of rib exposure area to the metal bed showed that the exposure area has a significant influence on heat transfer as well as the hydrogen absorption rate. Further, the main interest of this study was to numerically assess the impact of the compartmentalization of the reactor on heat transfer and hydrogen absorption. Simulations were carried out with vertical division (Case 2) and with both vertical and horizontal division (Case 3) of the reactor. The results showed that the compartmentalization had a substantial influence on heat removal from the bed, which significantly accelerated the hydrogen absorption. Further, the compartmentalization helped to maintain uniform temperature distribution, and as a consequence, uniform utilization of the metal powder, and eventually, uniform metal powder density in the reactor. Finally, to study the effect of the metal bed volume displaced by the compartment structures, two more simulations were carried out by considering the compensation volume from the expansion volume. The results proved that for the reactor geometry considered in this study, the heat transfer was enhanced by having the compartment structures negate the effect of the metal bed volume replaced by them. This paper contributes to enhancing the fundamental understanding of heat and mass transfer in an MH tank, and in particular, explores the stack reactor configuration for hydrogen storage. Most widely studied inner tube exchangers have proved to be effective for heat transfer from/to the metal bed; however, they are also very susceptible to the leaking cooling/heating medium due to tube erosion or stresses developing in the metal bed. With the present design of the reactor, such issues can be eliminated.

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